Caio Valerio Goulart Correia

AI Specialist - Full-Stack Developer & DevOps Engineer | Industrial Automation Specialist | Mechanical Engineering Student | Node.js • Python • AWS • TypeScript

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Backend engineer with 9+ years of technical experience building and maintaining APIs, services, and developer-facing tools — primarily TypeScript and Node.js, with deep Python alongside. Shipped and operates a production HR/operations platform (200+ employees) whose Node.js/TypeScript backend exposes 600+ API routes secured by JWT, two-factor authentication, and role-based authorization (PostgreSQL RLS + ACL). Defines cross-product API contracts (HMAC-authenticated sync), hardens third-party integrations with retries and fallback on rate-limit/503 responses, and maintains open-source developer tools on GitHub. Also deep in AI: LLM training/evaluation (SFT, DPO/RLHF), local serving (vLLM, SGLang), and RAG. Fluent English (C1), remote-first with international teams.

SKILLS

TypeScript · Node.js · Express · Next.js · REST APIs · API Design · GraphQL · JWT & Auth · Python · FastAPI · PostgreSQL · Supabase · PostGIS · Docker · Kubernetes · AWS · GitHub Actions · CI/CD · Linux · Git · Java · Go · Generative AI · LLMs · RAG · Model Fine-Tuning · MLOps · vLLM · MCP · Backend Development · Data Engineering · Security · Compliance

PROFESSIONAL EXPERIENCE

System Analyst

ABZ SERVIÇOS LTDA · Brazil, On-site · Mar 2025 – Present

- Architected and shipped EmployeeHub (portal.groupabz.com), an HR/operations portal used by 200+ employees — personnel records, crew management, e-Social, vacation, reimbursement, and ACL — on Next.js 15, TypeScript, and Supabase (PostgreSQL with Row-Level Security) on Vercel; reduced HR manual workload by ~40% and approval time by ~95%.
- Built and maintains the platform's Node.js/TypeScript backend — 600+ REST API routes (Next.js App Router) with JWT authentication, two-factor authentication (2FA), and role-based authorization via PostgreSQL Row-Level Security and ACL, protecting sensitive HR/PII data under LGPD.
- Delivered the ABZ product suite around the portal: PontoFlow (bilingual offshore timesheets with manager approval, field annotations, and deadline locks enforced in UI and RLS), Invoice ABZ (FastAPI/SQLAlchemy engine turning payroll and charge-rate spreadsheets into client invoices), and Ticket-Manager (support automation via Microsoft Graph).
- Defined the cross-product API contract (HMAC-authenticated import/export) between the portal and the PontoFlow timesheet product, keeping collaborator identity and period behavior consistent across both systems.
- Led the cloud migration of legacy on-premise Windows Servers (2012–2025), reaching 80% completion with zero production downtime.
- Implemented end-to-end CI/CD with GitHub Actions and Docker, administered Proxmox virtualization and AWS S3, and deployed a WireGuard VPN for 50+ remote employees — sharply reducing connectivity support tickets.

Owner / Director

SmartStore · Rio das Ostras, RJ · May 2021 – Present

- Founded and managed a technical sales and services company specializing in B2B industrial solutions, growing market presence by 30% and keeping customer satisfaction above 95%.
- Led negotiations, logistics and procurement of custom digital solutions, preparing complex technical proposals and using a CRM to track opportunities.
- Delivered end-to-end solutions that increased customer engagement and operational efficiency.
- Managed on-premises server, PC and network/firewall infrastructure, with proactive monitoring, troubleshooting and security best practices and hardening.
- Automated critical IT tasks, backups and hardware maintenance with Python and Shell scripting, including custom monitoring scripts, cutting manual intervention by 40%.
- Deployed custom Windows builds and configured ERP and software (Totvs RM, AutoCAD, Navisworks), standardizing deployments and documenting installations.
- Established contingency and disaster recovery plans.
- Produced 50+ bilingual technical manuals and system specifications for international clients.

AI Specialist (Contract)
micro1 · San Francisco County, CA, US, Remote · Aug 2026 – Sep 2026

- Trained and evaluated LLMs for frontier-model programs, running Supervised Fine-Tuning (SFT) and preference optimization (DPO/RLHF) with PyTorch, Hugging Face PEFT (LoRA/QLoRA), Unsloth, and Axolotl.
- Ran adversarial testing (red teaming) to surface hallucinations, logic flaws, and bias, feeding corrections back into training datasets.
- Reviewed and benchmarked AI-generated code across Python, TypeScript, Node.js, FastAPI, React, Next.js, and JavaFX — assessing data-structure choices, execution behavior, and complex API integrations.
- Deployed local LLM serving and agent harnesses on Windows/Linux (vLLM, SGLang, llama.cpp) on dedicated NVIDIA hardware (L4 GPU, 24GB VRAM), plus MCP tool servers for retrieval and graph tools; measured latency/throughput with custom evaluation harnesses.
- Curated high-quality training and synthetic datasets (Self-Instruct, Evol-Instruct) with automated validation for dataset integrity, schema compliance, and model alignment.

Technical Drafter II
BRtech3D Laser Scanning Solutions · Macaé, RJ, Hybrid · Mar 2024 – Mar 2025

- Executed field data collection using Leica TLS equipment and introduced Quality Assurance (QA) processes for outlier cleaning and multiple scan registration.
- Pioneered the implementation of Unreal Engine for visualization and analysis of digital twins of industrial facilities, with Leica Cyclone integration, interactive navigation and real-time analysis, photorealistic models for predictive maintenance and simulation, and Blueprint Visual Scripting and C++.
- Operated professional-grade 3D laser scanners to capture data from complex facilities, processing point clouds in Leica Cyclone and cleaning and optimizing them for real-time rendering.
- Designed technically detailed 3D models, AutoCAD drawings and as-built/BIM documentation for predictive maintenance, coordinating with engineering to ensure dimensional accuracy and automatically detecting material deviations from point clouds.
- Led R&D efforts to optimize point cloud processing workflows, automated reporting and custom tooling; the automation pipelines cut processing time by 25%, and the technical documentation standardized the processes.
- Collaborated with multidisciplinary teams on the continuous improvement of capture and analysis methodologies.

Python Developer
BRTech3D · Mar 2024 – Mar 2025

- Developed high-performance Python scripts (NumPy, SciPy, Open3D) to automate geospatial data pipelines, optimizing the processing of dense point clouds (more than 100 million data points).
- Implemented data routines that increased processing speed by 40% and improved data compression, generating significant savings in time and resources across dozens of projects.
- Modeled and optimized PostgreSQL databases using the PostGIS extension for complex spatial queries and seamless integration via REST APIs.

Laser Scanner Operator III
Master Serviços · Macaé, RJ, On-site · Oct 2023 – Mar 2024

- Led field scanning and point cloud capture projects at complex industrial facilities, increasing accuracy by 20% through capture optimization and reducing completion time by 15% through workflow refinement.
- Produced isometric and orthogonal drawings, material deviation analyses and assembly simulations, with automated quality checks and technical feedback to mechanical teams that delivered 15% higher efficiency and 10% fewer errors.
- Mentored new operators in data capture and processing techniques.
- Operated high-precision 3D laser scanning equipment (Leica) in complex industrial and archaeological environments.
- Led field campaign planning and technical documentation, ensuring millimeter-level precision and data backup redundancy for large-scale data collection.

Automation Technician
Jackkk Connection Comercial · Macaé, RJ · Apr 2021 – Apr 2022

- Delivered specialized technical training in hydraulics, pneumatics and PLC programming, with structured educational modules and hands-on training for customers and partners.
- Designed and simulated complex automation systems with FluidSim and Automation Studio, validating them before implementation to reduce errors and rework.
- Managed technical sales, import logistics and specialized B2B support, including negotiation with international suppliers and technical after-sales support.
- Created system documentation and operating guides for customers.

Training Specialist
Jackkk Connection Comercial · May 2017 – Mar 2021

- Coordinated and delivered technical training for over 500 professionals in PLCs, hydraulics, and industrial automation.
- Developed educational simulations and managed B2B technical sales, building direct relationships with international suppliers.

Partner / Systems Analyst
TechFix reparos e manutenção · Rio das Ostras · Feb 2018 – Dec 2018

- Co-managed the business as a partner, serving clients in equipment and systems repair and maintenance.
- Analyzed and configured client systems and computers, handling fault diagnosis, installation and technical support.

Industrial Automation Technical Assistant
Jackkk Comercial onshore e offshore · Granja dos Cavaleiros (Macaé) · Jun 2017 – Dec 2017

- Supported assembly, testing and maintenance of hydraulic, pneumatic and PLC automation systems for onshore and offshore clients.
- Simulated hydraulic and pneumatic circuits in FluidSim and Automation Studio to validate solutions before implementation, applying Parker and Festo hydraulics and pneumatics training.

Technical Team Lead
Diesel Line Cambuí · Aug 2016 – May 2017

- Led field teams specialized in maintenance of large engines and turbines (CAT, Cummins, MAK).
- Managed technical support tickets in English for international clients and formally documented failure analysis processes.

Product Support
Diesel Line Cambuí · Rio das Ostras · Aug 2015 – May 2017

- Created technical data books and specialized documentation.
- Managed bilingual negotiations with customers.
- Led teams supporting engine and turbine repairs (CAT, MAK, Cummins, Wärtsilä, STX).

Technician
Diesel Line Cambuí · Oct 2015 – Aug 2016

- Started career supporting commercial processes, CRM management, and strict application of compliance procedures for diesel and gas systems maintenance.

EDUCATION

B.S. in Mechanical Engineering (in progress)
Centro Universitário UniFatecie · May 2024 – May 2027 (expected)

High School, Industrial Automation
Instituto Federal Fluminense

High School
Alternativa Colégio · 2013 – 2015

Technical Diploma in Industrial Automation
ETP Escola Técnica · 2013 – 2015

CERTIFICATIONS

Micro1 Hackaton 2026
micro1 · Jul 2026

ChatGPT
UniFatecie · Aug 2024 · ID 332515

Fundamentos da Inteligência Artificial
UniFatecie · Jul 2024 · ID 332515

Hidráulica e Pneumática
Parker Hannifin · Jul 2017

Hidráulica e Pneumática
Festo · Jul 2017

Técnico de Automação
Crea-RJ · Apr 2015 · ID 2017115401

RELEVANT PROJECTS

Painel ABZ
Next.js 15 · TypeScript · PostgreSQL · Supabase
Corporate operations portal for ABZ Group: HR, offshore crew management, payroll, reimbursements and internal comms on Next.js 15, React 18 and Supabase/PostgreSQL (50+ tables with RLS) behind a Node.js/TypeScript API layer of 600+ routes with JWT auth and 2FA. Ships e-Social (13 events, RSA-SHA256 XML, SOAP mTLS), OCR (Tesseract) for ASO/ID docs, digital signatures, MIO ERP sync, and an AI companion with LiveKit voice, MS Graph and autonomous KPI agents.
https://github.com/Caiolinooo/EmployeeHub

PontoFlow
Next.js 15 · TypeScript · Node.js · Supabase
Bilingual (pt-BR/en-GB) timesheet system for offshore crews with manager approval, field-level annotations, and monthly deadline locks enforced in UI and Postgres RLS. Node.js server runtime with Nodemailer cron reminders and multi-tenant admin; syncs with EmployeeHub through an HMAC-authenticated import/export API contract.
https://github.com/Caiolinooo/PontoFlow

CloudSec & FinOps Auditor
Next.js · TypeScript · RAG · Gemini
Compliance auditor that evaluates cloud architecture scenarios against CIS / SOC 2 / FinOps-style policies, returning structured risk, cost impact, citations, and remediation steps. Hybrid RAG (BM25 + TF-IDF + reciprocal rank fusion) over versioned policies; resilient LLM calls with automatic retry and model fallback on 503/rate-limit responses; Vitest tests and CI evaluation pipeline.
https://github.com/Caiolinooo/cloudsec-finops-auditor

SGLang Commander
Python · FastAPI · React · PySide6
Open-source desktop (PySide6) and web (React + FastAPI) console for SGLang inference servers. Starts and stops servers, streams multimodal chat (image/audio), plots live GPU/VRAM/latency metrics, deploys Hugging Face models in one click, runs P50/P95/P99 benchmarks, and supports Docker plus ZeroTier remote deploys.
https://github.com/Caiolinooo/sglang-commander

Visualizador 3D FARO
Three.js · React · Open3D · MCP
Matterport-style web viewer for FARO Focus scans. Renders PTS/E57 point clouds and TrueView 360 panoramas with Three.js and React Three Fiber, plus dollhouse view, floor plans, distance measurement, annotations, auto-tours and a Model Context Protocol bridge so local LLMs can drive the 3D scene.
https://github.com/Caiolinooo/Visualizador3D_Matterport_Clone

Gerador de Malha Open3D
Python · Open3D · NumPy
Desktop pipeline that turns industrial point clouds into production meshes: Poisson reconstruction, hole filling, density-based vertex cleanup, smoothing and export to PLY/OBJ/STL — built for geospatial and laser-scan workflows with 100M+ points.
https://github.com/Caiolinooo/Projeto_Open_Mesh_

AD Migration Suite
C# · .NET 8 · WPF
Enterprise Active Directory migration from Windows Server 2012/2016 to 2019/2022. Agent-based architecture on a single port (8765) avoids WinRM/Kerberos setup, preserves SID history, replicates GPOs/OUs and migrates file shares with ACL translation across domains.
https://github.com/Caiolinooo/AD_Migrator

Auditoria de Dados
Python · Pandas · Dash · Plotly
Server data-audit toolkit with parallel directory scans, file-type and growth analysis, caching and an interactive Dash/Plotly dashboard. Generates timestamped HTML and XLSX reports with client-level filters for large file-server inventories.
https://github.com/Caiolinooo/Python_Audit_Script

Network Monitor
Python · Flask · PWA
Cross-platform network monitor (Windows, Linux, macOS, including Windows Server 2012) with live download/upload/latency, speed tests, history charts, PDF reports and PWA install for offline use on the LAN.
https://github.com/Caiolinooo/network-monitor

TECHNICAL SKILLS

AI & LLM Engineering
Generative AI, LLMs, RAG, Prompt Engineering, Model Fine-Tuning (SFT, DPO/RLHF, LoRA/QLoRA), MLOps, vLLM, SGLang, llama.cpp, MCP.

Backend & APIs
TypeScript, Node.js (Next.js server runtime, Express, REST API routes), Python (FastAPI, SQLAlchemy), API design & cross-product API contracts, authentication & authorization (JWT, 2FA, RLS, ACL, HMAC), resilience patterns (retries, fallback, rate-limit/503 handling), GraphQL, Java (Spring Boot), Go.

Cloud & DevOps
Linux, Git, DevOps, Docker, Kubernetes, AWS (S3), GitHub Actions (CI/CD), Vercel, Cloudflare, Proxmox, WireGuard VPN, Terraform.

Databases
PostgreSQL, PostGIS, Supabase (Row-Level Security), SQLite (FTS5), NoSQL, ETL/data pipelines.

Governance, Security & Compliance
Data Governance, Security, Compliance, LGPD/PII controls.

LANGUAGES

Portuguese: Native | English: Fluent (C1) | Spanish/Italian: Advanced technical reading.